PH 142: Introduction to Probability and Statistics in Biology and Public Health

Summer 2026

Summer Session D, July 6–August 14, 2026

4 semester credits

Please note: This syllabus is subject to change. The course website will contain the most current schedule, links, and assignment information: https://ph142-ucb.github.io/su26/.

Course Format and Meeting Information

Course	PBHLTH 142, Section 001, Lecture 001
Class number	13581
Instruction mode	Online
Lecture	Monday through Friday, 9:30–10:59 AM Pacific, Internet/Online. Recordings will be posted daily; live Zoom lectures will be held every other day.
Labs	Online and synchronous, Monday through Thursday
Discussion platform	Ed Discussion. We will use Ed Discussion, not Piazza, for course questions and announcements.

Lab Sections

Section	Class #	Days	Time	Place
101A	13583	Mo, Tu, We, Th	11:00–11:59 AM	Internet/Online
102A	13584	Mo, Tu, We, Th	11:00–11:59 AM	Internet/Online
103A	13585	Mo, Tu, We, Th	5:00–5:59 PM	Internet/Online
104A	13586	Mo, Tu, We, Th	5:00–5:59 PM	Internet/Online

Course Description

This course is an introduction to statistics and data science, primarily for MPH and undergraduate public health majors, and others interested in public health topics. The course can be divided into three parts.

Part I focuses on using R to explore datasets and summarize univariate and bivariate distributions visually and numerically. We will also introduce concepts around sampling and how study data are acquired. Specifically, we will use the `dplyr` and `ggplot2` packages to manipulate and visualize data sets in R.

Part II introduces classical problems in probability and the Normal, Binomial, and Poisson distributions. We also introduce concepts around sampling variability and the Central Limit Theorem.

Part III introduces statistical inference: the process of estimating statistics from samples to make inference about populations. During all parts of the course we will use real and simulated data sets to gain experience conducting biostatistical analyses using R. We will follow the PPDAC model: Problem, Plan, Data, Analysis, and Conclusion.

Prerequisites

No specific coursework is required, and no programming background is necessary. We expect a level of math equivalent to a US secondary school algebra course.

Course Learning Objectives

After successfully completing Part I of the course, you will be able to:

- Extract relevant statistical information from published articles in the scientific and popular press.
- Describe distributions of variables visually and calculate summary statistics for centrality and spread.
- Determine appropriate graphics for distributions and provide R code to manipulate and visualize data frames.
- Identify basic sampling strategies and study designs used in public health.
- Describe core concepts of ethics in public health.
- Perform basic data manipulation in R.

After successfully completing Part II of the course, you will be able to:

- Compute probabilities using general probability rules.
- Understand statistical independence.
- Identify and describe Binomial and Poisson random variables.
- Compute probabilities using basic properties of the Normal distribution.
- Express epidemiologic measures as probabilities.
- Describe sampling variability and the Central Limit Theorem.
- Write R code to compute probabilities for the Normal, Binomial, and Poisson distributions.

After successfully completing Part III of the course, you will be able to:

- Estimate means, proportions, and differences between means and proportions; compute confidence intervals; and perform statistical tests.
- Choose appropriate measures and statistical tests for associations between variables.
- State the assumptions and importance of assumptions for statistical tests.
- Describe and check assumptions for simple linear regression.
- Interpret confidence intervals and statistical tests for regression intercept and slope coefficients.
- Write R code snippets to generate confidence intervals, perform hypothesis tests, and calculate p-values.
- Formulate a clear statistical question, address it with an appropriate method, and accurately report findings.

Instructor Information and Communication

Instructor: Jiaxin Qing (he/him/his)
Instructor email: jxqing@berkeley.edu
Instructor availability: Please book through email.
Course email for non-content inquiries: 142gsi@berkeley.edu

While the instructor will interact with the whole class and oversee activities and grading, Graduate Student Instructors (GSIs) will be your main point of contact for assignment and course requirement questions. GSIs will lead lab sections, hold office hours, and facilitate discussion.

All questions about course material should be asked on Ed Discussion. We will not answer course-content questions by email. This keeps answers available to the full class. For questions and concerns not related to course content, email 142gsi@berkeley.edu. Do not use the bCourses emailing system for course communication.

Office Hours

GSI office hours will be posted on the course calendar. You may attend the office hours of any GSI regardless of your lab section. Instructor office hours will primarily be offered by appointment through email.

Students with Disabilities

If you are eligible for disability-related accommodations through UC Berkeley's Disabled Students' Program, please send us your letter of accommodation through the DSP student portal as soon as possible, so we can ensure your accommodations are met.

If you have any questions about your accommodations for PH142, please email 142gsi@berkeley.edu. Do not directly email any medical documentation to the PH142 teaching team. If your accommodation status changes at any point during the semester, please send us an updated letter of accommodation through the DSP student portal.

If you believe you may be eligible for disability-related accommodations, please visit the Disabled Students' Program website to begin an application as soon as possible. The application process may take several weeks.

Course Materials and Technical Requirements

Course website: https://ph142-ucb.github.io/su26/. The course website will contain the syllabus, assignment descriptions, course resources, the most up-to-date schedule, and assignment information.

We will use R, RStudio, and Datahub. Use of R, RStudio, and Datahub is required for homework assignments and lab exercises and requires an internet connection and web browser. You will learn how to use R, RStudio, and Datahub during the first week of classes.

Optional textbook: *The Practice of Statistics in the Life Sciences* by Brigitte Baldi and David S. Moore. The 4th edition is the latest one, but previous editions are fine. The textbook is not required; course materials take precedence where there are differences.

Learning Activities and Assignments

All times listed are Pacific Time.

Lectures

Lectures meet Monday through Friday from 9:30 to 10:59 AM Pacific. Lectures are online. Lecture recordings will be made available every day. Live Zoom lectures will be held every other day; on non-live days, students should use the posted recordings and materials to keep pace with the course.

Lab/Discussion Sections

Lab/discussion sections meet synchronously online Monday through Thursday. Sections 101A and 102A meet 11:00–11:59 AM; sections 103A and 104A meet 5:00–5:59 PM. Lab attendance will be recorded for sessions with lab assignments. Zoom meeting data will be used to track your attendance; you must log into the lab meeting with your berkeley.edu email address. Attending 8 out of 11 labs will earn full points for this portion of your participation grade. Lab sections will include hands-on coding assignments in R, review of key concepts, group activities, and support for group data projects.

Quizzes

Daily quizzes are intented to test concepts learned during that day's lecture. Quizzes are administered via Gradescope and are available for 24 hours. Quizzes are released at 12:00AM and due by 11:59PM. Once opened, students have 30 minutes to complete the quiz. There are 26 quizzes total.

Lab Assignments

Lab assignments are intended to practice lecture concepts in a practical programming environment. They are intended to be completed during lab section with support from the GSIs. Lab assignments are released at 12:00AM and due by 11:59PM. Lab exercisesare submitted through Datahub to Gradescope and are graded for correct completion. There are 11 lab assignments total.

Midterms

There will be three midterm exams throughout the term. The midterms will take place on July 16th, July 30th, and August 13th. The midterm exams are administered via Gradescope and will be open for 24 hours. Once opened, you have 1 hour to complete the exam. Exams may cover material presented in lecture, supplemental videos, discussion, lab sections, and R coding syntax unless otherwise noted.

Students may use one page of handwritten or printed notes for the midterm exams. You may not use the internet to search for answers or inform your answers during exams. You may not use AI tools during the midterm exams. While taking an exam, you are prohibited from discussing the exam with anyone other than the PH 142 instructional team. Evidence of academic misconduct may result in a zero on the exam or further disciplinary action.

Data Skills Demonstration Group Project

The purpose of the group project is to use public health or biological data that you find or have access to and use it to demonstrate statistical concepts learned throughout the course. Data project groups are randomly assigned by the teaching team and will be released on Tuesday, July 7th during lab section. Students are required to check in with their lab GSI prior to submitting each part of the data project for participation credit.

The project will be submitted in three parts. The Summer 2026 deadlines are:

- Part I: July 17 at 11:59 PM Pacific.
- Part II: July 31 at 11:59 PM Pacific.
- Part III: August 14 at 11:59 PM Pacific.

Participation

Participation is worth 10% of your overall course grade. Your participation grade consists of lab attendance (7%) and three data project GSI check-ins (3%). You will earn full credit for the lab attendance portion by attending at least 8 of the 11 lab sections with lab assignments.

Optional Homework Assignments

Homework problem sets are optional and not submitted for credit. These assignments provide practice and preparation for exams, labs, and the data project. Solutions will be posted after students have had time to complete the assignment.

Grading

The final course grade percentages are:

Category	Percentage	Notes
Quizzes	10%	There are 26 quizzes total
Lab Assignments	10%	There are 11 lab assignments total
Midterm I	15%	
Midterm II	15%	
Midterm III	20%	
Data project	20%	Submitted in three parts
Participation	10%	Includes lab section attendnace (7%) and data project check-ins with your GSI (3%)

Tentative Course Schedule

The schedule below is tentative. The website is the authoritative schedule.

Week	Dates	Topics and major deadlines
1	Jul 6–10	Course introduction; PPDAC; Datahub; working with data in R; visualization; numerical summaries; relationships between variables. Labs released daily and due the same day; Weekly Assignment 1 released.
2	Jul 13–17	Regression basics; two-way tables; samples and observational studies; study design and ethics; introduction to probability. Labs released daily and due the same day; Weekly Assignment 2 released; group confirmation due Sunday.
3	Jul 20–24	Probability rules; independence; Normal, Binomial, and Poisson distributions; sampling distributions and the Central Limit Theorem. Midterm review; Weekly Assignment 3 released; Part I due Sunday.
4	Jul 27–31	Online midterm; confidence intervals; hypothesis testing; power; from z to t tests. Weekly Assignment 4 released.
5	Aug 3–7	Comparing two means; paired t tests; ANOVA; non-parametric tests; inference for proportions; inference for linear regression. Weekly Assignment 5 released; Part II due Sunday.
6	Aug 10–14	Linear regression part II; chi-square goodness of fit; chi-square independence; final review; Weekly Assignment 6 released; online final exam; Data Project Part III due Friday.

Due Dates, Late Work, and Regrades

Due dates for all graded assignments will be listed on the course website and communicated in announcements on Ed Discussion.

Late work. Assignments submitted within 24 hours after the due date will be accepted with a 50% penalty. Extensions can be made for DSP students and should be requested by emailing the course email. Other extension requests should explain the situation and must include documentation.

Regrades. Regrade requests must be submitted through Gradescope within three days after grades are released. Regrades are for correcting an error in the application of the published rubric. If you request reconsideration of one part of an assignment, instructors may reconsider grades on the entire assignment. Due to the short turnaround time for final grade submissions, we generally cannot accommodate regrade requests for Midterm III.

Academic Integrity

You are a member of an academic community at one of the world's leading research universities. Any test, paper, code, or report submitted by you and bearing your name is presumed to be your own original work unless otherwise specified. You may use words or ideas from other individuals, publications, websites, or other sources only with proper attribution.

Students may not circulate or post course materials, including handouts, exams, syllabi, assignments, or solutions, without written permission from the instructor. If you are unclear about expectations for completing an assignment or taking an exam, seek clarification from the instructor or a GSI beforehand.

Strategies for Successful Learning

Based on past experience with this course, we recommend that you:

- Attend lab consistently.
- Ask questions in lecture, lab, office hours, and Ed Discussion.
- Study in a group and take turns explaining concepts to each other.
- Review questions and concepts from weekly assignments and exams where you lost points.
- Use practice exams to become familiar with question types and to practice the material.

Take Care of Yourself

Do your best to maintain a healthy lifestyle during this intensive summer session by eating well, exercising, getting enough sleep, and taking time to recharge your mental health. If you start to feel overwhelmed, be kind to yourself and reach out for support. Seeking help is a courageous thing to do for yourself and for those who care about you.

Support resources include emotional, physical, safety, social, and basic wellbeing resources for students. Academic resources can be found at the Student Learning Center and English Language Resource sites. If you notice that a classmate may need assistance, please use campus resources or reach out to the GSIs or instructor.

Incomplete Course Grade and Course Evaluation

Students who have substantially completed the course but, for serious extenuating circumstances, are unable to complete remaining course activities may request an Incomplete grade. This request must be submitted in writing to the GSI and instructor and must include verifiable documentation.

Before the course ends, please participate in the course evaluation. Your feedback helps improve the course and future instruction.